

1. Use the **limit of Riemann sum definition** of the definite integral to evaluate $\int_0^2 (4 + 3x^2) dx$.

Note: You may use right-endpoint Riemann sum. No credit will be given if you use other method.

Basic Summation Formulas: $\sum_{i=1}^n c = cn$, $\sum_{i=1}^n i = \frac{n(n+1)}{2}$, $\sum_{i=1}^n i^2 = \frac{2n^3 + 3n^2 + n}{6}$

$$\sum_{i=1}^n i^3 = \frac{n^4 + 2n^3 + n^2}{4}$$

$$\Delta x = \frac{2}{n}$$

$$x_i = \frac{2i}{n}$$

$$\begin{aligned} f(x_i) &= 4 + 3\left(\frac{2i}{n}\right)^2 \\ &= 4 + 3 \cdot \frac{4i^2}{n^2} \\ &= 4 + \frac{12}{n^2} i^2 \end{aligned}$$

$$R_n = \sum_{i=1}^n f(x_i) \Delta x$$

$$= \sum_{i=1}^n \left(4 + \frac{12}{n^2} i^2\right) \frac{2}{n}$$

$$= \sum_{i=1}^n \left(\frac{8}{n} + \frac{24}{n^3} i^2\right)$$

$$= \left(\sum_{i=1}^n \frac{8}{n}\right) + \frac{24}{n^3} \left(\sum_{i=1}^n i^2\right)$$

$$= \frac{8}{n} \cdot n + \frac{24}{n^3} \cdot \frac{2n^3 + 3n^2 + n}{6}$$

$$= 8 + 4 \left(2 + \frac{3}{n} + \frac{1}{n^2}\right)$$

$$= 8 + 8 + \frac{12}{n} + \frac{4}{n^2}$$

$$= 16 + \frac{12}{n} + \frac{4}{n^2}$$

$$\int_0^2 (4 + 3x^2) dx = \lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} \left(16 + \frac{12}{n} + \frac{4}{n^2}\right) = 16$$

2. Use Part 1 of the Fundamental Theorem of Calculus to find the derivative of $F(x) = \int_0^{\ln x} \sqrt{t^4 + 1} dt$.

$$F'(x) = \sqrt{(\ln x)^4 + 1} \cdot \frac{1}{x}$$